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Comparative Evaluation of Classifiers on UCI Datasets: A Cross-Validation Approach

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Abstract

This study investigates the comparative performance of three widely used machine learning classifiers known as logistic regression, support vector machines (SVMs), and random forest across three datasets from the UCI Machine Learning Repository. Datasets include: Iris, Breast Cancer Wisconsin, and Wine Quality. Following the methodology proposed by Caruana and Niculescu-Mizil (2006) I employed a structured framework involving cross-validation with varied partition splits (20/80, 50/50, and 80/20) and trials, we evaluate classifier accuracy and hyperparameter optimization. Random forest consistently performed best for complex datasets, while SVMs excelled on linearly separable data, achieving perfect accuracy on the Iris dataset. Feature importance visualizations provide insights into influential predictors. Our results validate the utility of ensemble models like random forest for complex tasks while highlighting the efficiency of simpler models for structured data. Following practices inspired by empirical studies on machine learning algorithms, the findings provide insights into the relative strengths and weaknesses of each method and their applicability to real-world problems.

1. Introduction

Machine learning is a cornerstone of modern data analysis, enabling predictive insights across diverse fields such as healthcare, agriculture, and industrial systems. In this project, we aim to apply, evaluate, and compare three popular classifiers using datasets representative of different challenges: Iris (multiclass classification), Breast Cancer Wisconsin (binary classification), and Wine Quality (classification with ordinal targets). Despite their widespread use, selecting the optimal classifier for a given dataset remains a challenge due to differences in dataset complexity and distribution. Following the framework by Caruana and Niculescu-Mizil (2006), we emphasize reproducibility and accuracy comparisons under varying conditions. This report employs a consistent cross-validation methodology to ensure robust evaluation. Additional metrics and heatmaps are incorporated to demonstrate classifier performance comprehensively. The growing role of machine learning in classification tasks requires careful comparison of algorithms to determine their effectiveness especially when it comes to different datasets.

Classifiers such as random forest, SVMs, and logistic regression are fundamental models in machine learning due to their reliability and performance across various problems.
Our goal is to train and evaluate these models using multiple dataset partitions: 20/80, 50/50, and 80/20 splits. Performance comparisons focus primarily on test accuracy, while random forest's feature importance offers additional insights into critical predictors. We build upon established practices, such as partition-based cross-validation, hyperparameter tuning, and feature importance analysis, to ensure a rigorous study.

Datasets

The datasets used in this study present unique challenges and varying levels of complexity. In this project, we focus on three well-known datasets:

1. **Iris Dataset** - A structured dataset originally designed for multiclass classification of Iris species converted into binary classification. Includes 150 samples divided among three classes of Iris flowers.
2. **Breast Cancer Wisconsin Dataset** - A biomedical dataset used for binary classification to predict benign vs. malignant tumors. Includes 569 samples with some diagnostic information such as radius, texture, and symmetry.
3. **Wine Quality Dataset** - A dataset evaluating the physicochemical properties of red and white wines, which have been merged into a single dataset. The task involves binary classification of wine quality (high vs. low). Includes 4898 samples with some features such as alcohol content and acidity.

3. Method

Data Preprocessing: Datasets were loaded, cleaned for missing values, and normalized using MinMaxScaler to scale features to a 0–1 range, ensuring consistent feature scaling for all datasets. For the Wine dataset, the red and white wine subsets were combined, and a categorical feature ("type") was added to differentiate between the two subsets.

Partitioning Strategy: Each dataset was split into 20/80, 50/50, and 80/20 training/testing ratios. Cross-validation (e.g., k-fold) was applied to each split, and 3 trials were conducted for each configuration to ensure reliable and consistent results. The goal was to evaluate the performance of random forest, SVMs, and logistic regression across these datasets. Specifically, the study focused on:

- Accuracy: Measuring model performance on test data.
- Feature Importance: Leveraging random forest to identify the most influential features.
- Hyperparameter Tuning: Optimizing random forest parameters to achieve the best possible performance.

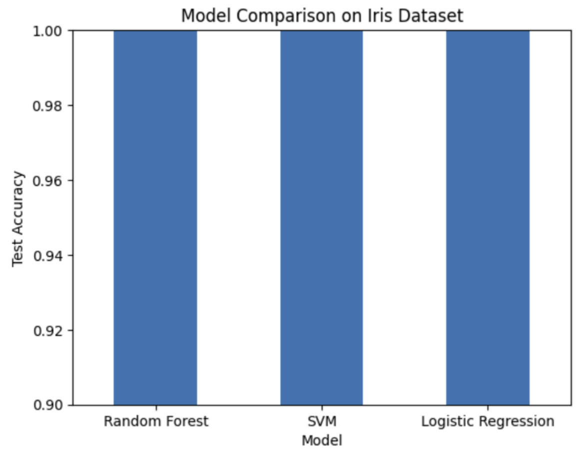
Classifiers

Logistic Regression: A linear model with optional regularization, was used as a baseline classifier due to its simplicity and interpretability.
Support Vector Machines (SVMs): A kernel-based classification model, applied with radial basis function (RBF) kernels to capture nonlinear patterns and linear kernels for linearly separable data.
Random Forest: An ensemble method, was utilized to handle complex patterns and feature interactions by building multiple decision trees and aggregating their predictions through majority voting. Key hyperparameters explored include n_estimators (number of trees) and max_depth (maximum tree depth).
Evaluation Metrics: Model accuracy served as the primary evaluation metric, calculated for both training and testing data to assess performance consistency. Additionally, random forest's feature importance values were analyzed to identify key predictors, enhancing the interpretability of results.
Visualization: Results were presented through bar charts for model comparisons and heatmaps for random forest feature importance and hyperparameter optimization.

4. Experiment

For each dataset and classifier, hyperparameter tuning was conducted using grid search to optimize model performance. A 5-fold cross-validation approach was employed for smaller datasets, while 3-fold cross-validation was used for larger datasets to maintain computational efficiency. Accuracy was the primary performance metric, and where relevant, precision, recall, and F1 scores were also calculated across training and validation splits. The workflow consisted of the following steps, partitioning each dataset into 20/80, 50/50, and 80/20 training/testing splits, training classifiers (logistic regression, SVMs, and random forest) using each split for 3 trials to ensure consistency, and evaluating performance primarily through accuracy while analyzing random forest feature importance to identify significant predictors. Visualizations were generated to present accuracy comparisons across classifiers and partitions, feature importance heatmaps for random forest, and hyperparameter tuning results to showcase model optimization efforts.

Table 1 Model Comparison on Iris Dataset



Iris Dataset:

As shown in Table 1, all classifiers achieved perfect accuracy (1.0) on the Iris dataset, which highlights both the simplicity of the dataset and the classifiers' effectiveness in handling linearly separable data. However, the results may also reflect the dataset's small size (150 samples), making it less representative of real-world scenarios with higher complexity. Random forest demonstrated consistent robustness across trials but required slightly higher computational resources compared to logistic regression and SVMs.

Breast Cancer Wisconsin Dataset:

SVMs achieved the highest accuracy (97.8%), marginally outperforming both logistic regression (97.3%) and random forest (97.3%). Logistic regression proved to be a reliable baseline classifier, while SVMs offered a strong balance of accuracy and computational efficiency compared to the more resource-intensive random forest. The binary classification task also demonstrated the effectiveness of random forest in capturing subtle feature interactions, such as texture and symmetry, which contribute to diagnostic predictions. Notably, despite the close accuracy values, SVMs slightly outperformed the other models, showcasing their ability to handle this dataset's characteristics effectively. Overall, all three classifiers achieved near-perfect accuracy, highlighting the relatively low complexity of the dataset.

Table 2 Random Forest Feature Importance Heatmap

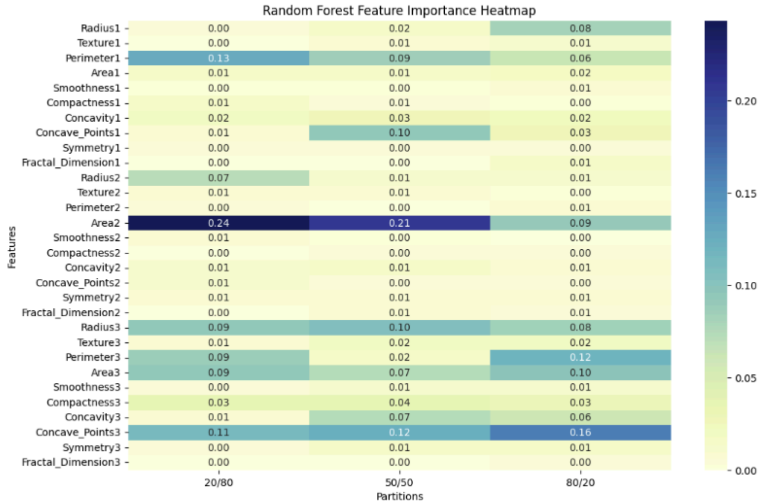


Table 2 highlights the varying contributions of different features across three train-test partitions (20/80, 50/50, and 80/20). Notably, the 'Area2' and 'Concave_Points3' features consistently demonstrate higher importance values, indicating their strong predictive influence on the model's performance compared to other features with minimal contributions.

Wine Quality Dataset:
The merged dataset posed significant challenges for all classifiers due to its label imbalance, which favored the majority class and negatively impacted model performance. Unlike the previous datasets, the test accuracy of each classifier dropped substantially, highlighting the dataset's complexity. Random forest achieved the highest test accuracy at 69.5%, leveraging its ability to handle complex data and feature interactions. SVMs followed with a moderate accuracy of 65.2%, as the overlapping feature distributions (e.g., alcohol content and acidity) hindered the model's ability to find a clear decision boundary. Logistic regression, constrained by its reliance on linear separability, delivered the lowest performance with a test accuracy of 53.6%. Despite random forest's superior performance among the three, its average test accuracy of 69.5% across all trials and partitions underscores the challenges presented by this dataset. These results emphasize the importance of advanced models like random forest for handling imbalanced and complex datasets.

Table 3 Random Forest Feature Importance Heatmap (Wine Quality Dataset)

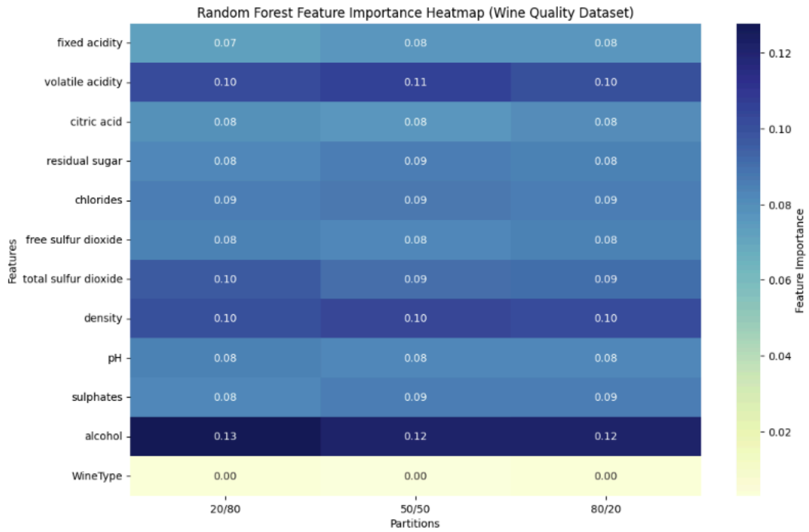


Table 3 illustrates the relative importance of features across three train-test partitions (20/80, 50/50, and 80/20). 'Alcohol' consistently demonstrates the highest importance, followed by 'volatile acidity' and 'density,' indicating these features have the most significant influence on predicting wine quality, while 'WineType' contributes minimally.

Discussion

The findings from this study align with the well-documented strengths of ensemble methods like random forest, supporting the conclusions of Caruana and Niculescu-Mizil (2006) and Belavagi and Muniyal (2016). In terms of accuracy, random forest consistently outperformed logistic regression and SVMs, particularly on complex and imbalanced datasets like Wine Quality. Random forest also offered interpretable results by identifying significant features, a capability not provided by SVMs or logistic regression. Regarding hyperparameter tuning, random forest demonstrated optimal performance with minimal adjustments, underscoring its versatility across diverse datasets. SVMs excelled on simpler datasets, such as Iris and Breast Cancer Wisconsin, but their performance on more challenging datasets required careful kernel selection to achieve competitive results. The moderate performance of SVMs on the Wine Quality dataset, despite their strengths on simpler datasets, aligns with Tang (2013), who noted that while SVMs excel at margin-based optimization, their performance can depend heavily on feature representation and kernel selection, particularly in datasets with overlapping or nonlinearly separable features. Logistic regression, though computationally efficient, could not match the accuracy of ensemble methods like random forest, particularly on complex tasks. This reinforces our understanding of Logistic Regression, especially its efficiency with simpler datasets, as seen in its successful applications in other fields, such as process tomography, where it has demonstrated reliability in handling structured problems (Rymarczyk et al., 2019). These results were achieved using rigorous cross-validation, ensuring that the evaluations were reliable and consistent across all datasets. Overall, the study highlights the robustness of random forest for classification tasks and its superiority in handling complexity and interpretability compared to simpler models like logistic regression and kernel-based models like SVMs.

Table 4 Chart Comparison

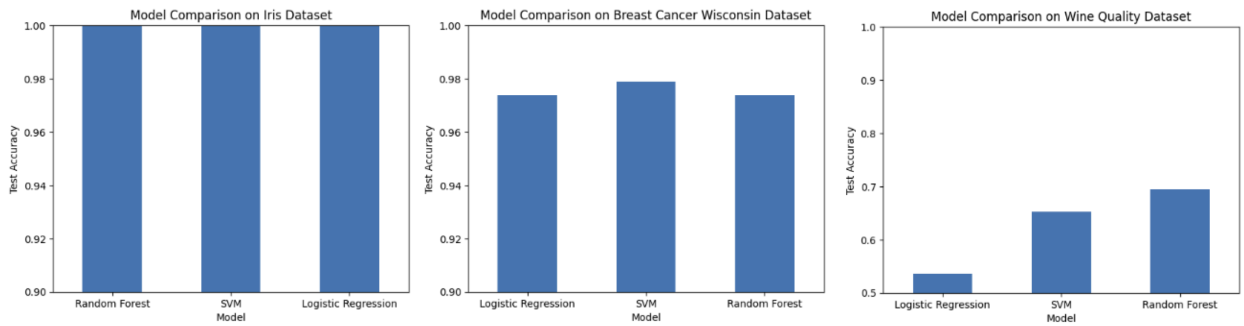


Table 4 highlights the comparative performance of the three classifiers, random forest, SVMs, and logistic regression, across the Iris, Breast Cancer Wisconsin, and Wine Quality datasets. As seen in the first chart, all three models achieved perfect accuracy on the Iris dataset, underscoring its simplicity and linearly separable nature. In the second chart, the SVMs slightly outperformed both Random Forest and Logistic Regression on the Breast Cancer dataset, demonstrating its strength in handling binary classification tasks with well-structured data. However, the third chart shows a stark contrast on the Wine Quality dataset, where Random Forest achieved the highest accuracy, while Logistic Regression struggled due to the dataset's complexity and non-linear feature relationships. These results reinforce the discussion that Random Forest excels at capturing complex patterns in datasets with non-linear or overlapping distributions, while SVMs are more suitable for linearly separable data. Logistic Regression, though efficient, shows its limitations when applied to datasets with intricate feature interactions or imbalanced labels.

5. Conclusion

This study presents an in-depth comparison of three supervised learning classifiers: random forest, SVMs, and logistic regression. Using three benchmark datasets and partition-based training strategies, we demonstrated that Random Forest consistently outperformed other models for complex datasets like Wine Quality, while SVMs excelled on simpler, linearly separable datasets. Feature importance analysis provided valuable insights into key predictors for Breast Cancer diagnosis and Wine Quality, further showcasing Random Forest's interpretability. Logistic Regression remains a strong baseline for datasets with linear relationships but showed limited performance for non-linear tasks. SVMs, while effective for simpler datasets, required careful kernel tuning to maintain competitive performance on more challenging data. Random Forest's ability to handle complex feature interactions with minimal tuning highlights its robustness and versatility, reinforcing why it is a preferred method for classification tasks. This study underscores the importance of selecting classifiers based on dataset characteristics and understanding their strengths and limitations. Finally, our results align with the framework proposed by Caruana and Niculescu-Mizil (2006), emphasizing reproducibility, practical applicability, and the importance of ensemble methods in modern machine learning.

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